SCI-BEAM

Formal Scientific/Engineering Contract

Effective PSF, Detector Calibration, Sensitivity, and Beammap APT

Scientific contract version v0.1
Document revision r0.3
2026-08-17

This is the normative, implementation-independent authority for the desired SCI-BEAM v0.1 estimator and products. It defines what a future implementation must realize and what evidence would be needed. It neither inspects nor claims conformance of current Citlali, existing APTs, tests, or production reductions.

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Abstract

SCI-BEAM fits standardized per-detector Beammaps in raw $\Delta f/f$ to measure an observation-local effective PSF core, reference-origin amplitude, and relative detector geometry; derives source-atmosphere-corrected, top-of-atmosphere nominal-beam `flxscale`; derives NEFD-like `sens` from detector-specific off-source scans; and publishes a mandatory scientific Beammap APT with independent quantity states and dense companions. The fitted Gaussian is a full 2-D effective-core summary, not an automatic intrinsic or complete PSF. Source atmosphere belongs to BEAM; target atmosphere remains downstream. Exact production thresholds and several sensitivity, adequacy, geometry, and kernel policies remain open.

Contents

1 Conformance object and prohibited inferences

The conformance object is one complete SCI-BEAM result for one immutable Beammap observation, including every attempted detector, the standardized input maps, fit and diagnostic state, mandatory APT, dense companions, and complete scientific lineage. A table of finite parameters alone is not a conformance object.

Conformance is evaluated against the shared notation, definitions, equations, assumptions, numbered requirements, and predictions below. The contract does not prescribe current classes, files, columns, algorithms, storage layout, or configuration names. It shall not be reconciled to historical behavior during authorship. A later audit may map each requirement to an implementation and evidence, but that mapping cannot revise the scientific meaning.

The strongest claim available from this document alone is algebraic contract correctness. It does not establish implementation conformance, parent-map or response fidelity, observational bias/coverage, science-impact qualification, or production readiness.

2 Logical input records

Table 1: Minimum logical inputs; representation is not prescribed.

Input	Required scientific content	Fail-closed conditions
Raw-signal convention	$\Delta f/f$ definition, reference resonance frequency, sign, zero/baseline, filtering/resampling/flagging, loading/tune/signal domain	Missing or incompatible physical observable; calibrated signal substituted
Standardized detector Beammap	stable observation and detector identity, $\Delta f/f$ pixels, valid support, map extent, parent mapmaking, pixelization, projection, gridding, filtering, response lineage, and noise/covariance state	Calibrated map; missing identity, WCS, support, or response provenance
Coordinate authority	full WCS or bounded local Jacobian, orthonormal tangent-plane frame/unit, scale, handedness, rotation, shear, azimuth metric, fitted-centroid sign/frame transformation, uncertainty	Raw azimuth/elevation treated Euclidean without metric; centroid-to-detector sign inferred from implementation; material unbounded linearization
Calibrator model	source identity/epoch, passband/spectrum, finite-source shape, zero-first-moment reference origin, uncertainty	Reference-origin or finite-source convention unresolved
Nominal calibration reference	fixed versioned nominal-beam tensor and $F^{\mathrm{TOA,nom}}_{s,a}$ in TOA $\mathrm{mJy\,beam}^{-1}_{\mathrm{nom}}$ under the same reference-origin convention	Unit, nominal-beam identity, or reference-origin convention missing
Source atmosphere	operator, inputs, support/weighting, effective transmission/correction, uncertainty/correlation	Missing source-atmosphere authority for `flxscale`; target atmosphere substituted
Field rotation	law/version, parent-sample time/elevation support and contribution weights propagated through fit support, conventional pivot/origin classification, uncertainty	Pixel fit weights substituted for parent timing; conventional pivot labeled physical; support inconsistent with fit
Soft prior	producer/version, coordinate/unit, identity scope, compatibility, reliability, gate, fallback	Exact truth or objective penalty inferred; blind route absent

3 Logical result and state records

Table 2: Required logical result records.

Record	Minimum contents	Primary evidence
Requested, effective, and realized state	model, source, support, background, covariance, prior, iteration, atmosphere, sensitivity, QC, and product policy as distinct one-way states	immutable resolution and no hidden defaults
Per-detector fit	amplitude, centroid, full effective tensor, FWHM, conditional area, background, residual, support, model Jacobian, joint map-fit covariance, convergence, prior route	Equations 9–12 and applicable predictions
Relative geometry	centroid-to-detector sign/frame transformation, raw and horizon coordinates, coordinate covariance, parent-sample support/contribution weights, effective derotation angle/range, rotation convention, origin/pivot state, rotated tensor	Equations 16–18
Response interpretation	effective-core state, empirical-map state, adequacy, measured-wing constraint, complete-PSF state, broadening tensor and PSD disposition	Equations 14–15
Calibration	observed and TOA-equivalent amplitude, required nominal-beam source flux, source-atmosphere route, `flxscale`, separate derived-product Jacobian/covariance, internal candidate/acceptance state	Equations 13 and 20
Sensitivity	detector trajectory classification, admitted scans, scan-level statistics, TOA correction, robust summary/scatter, positive `sens`, separate derived-product Jacobian/covariance, policy/uncertainty/limitations	Equations 13 and 24; open-policy gating
Independent state vector	separate availability, validity, uncertainty, and causes for every required quantity and use	no all-or-nothing detector collapse
Canonical lineage	all scientific identities, versions, parents, response/coordinate/atmosphere/calibration relations, content digests, limitations	deterministic reconstruction without row-order inference

4 Canonical Beammap APT scientific content

The mandatory APT is the compact scientific index into the complete BEAM result. A storage implementation may serialize equivalent tensor or identity representations, but shall preserve the following meanings and resolvability.

Table 3: Minimum per-detector canonical content.

Quantity	Meaning	Unit/state
Detector identity	stable detector occurrence and array/network/channel relation	identity
$(x_t, y_t)_{\mathrm{raw}}$	fitted relative coordinate after declared source-centroid sign/frame transformation	arcsec
(x_t, y_t)	horizon-derotated relative focal-plane coordinate	arcsec

Quantity	Meaning	Unit/state
Effective derotation	parent-sample contribution support/weights propagated through fit support, angle/range/rule/version	angular unit/state
Coordinate covariance	joint uncertainty of relative position	arcsec^2
Effective PSF-core tensor	authoritative complete 2-D core shape	arcsec^2
FWHM/orientation	compact shape; angle unavailable at circularity	arcsec/angular unit
A_d^{obs}	reference-origin observed-plane source excess	$\Delta f/f$
Amplitude uncertainty	conditional/total status with covariance identity	$\Delta f/f$
`flxscale`	absolute TOA nominal-beam conversion	$\mathrm{mJy\ beam}_{\mathrm{nom}}^{-1}/(\Delta f/f)$
`sens`	finite strictly positive TOA nominal-beam NEFD-like sensitivity using |`flxscale`|	$\mathrm{mJy\ beam}_{\mathrm{nom}}^{-1}\sqrt{\mathrm{s}}$
Independent states	coordinates, shape, angle, amplitude, calibration, sensitivity, adequacy, wings, health/kernel use	typed availability/validity/causes
`responsivity`	deprecated; noncanonical compatibility metadata only	no authority

APT-level lineage shall resolve observation/calibrator/epoch, source model and uncertainty, fixed nominal beam, TOA amplitude convention, passband/spectrum, source atmosphere, raw-signal definition, WCS/Jacobian/frame, centroid-to-detector sign transformation, parent-sample contribution weights, rotation law, conventional pivot status, common-origin rule, parent map response, PSF model, sensitivity estimator, contract/document versions, parents, and content digests.

The APT shall link immutably to empirical detector Beammaps, support maps, residuals, fit and derived-product Jacobians/covariances, backgrounds, adequacy diagnostics, broadening/wing state, off-source scan membership/statistics, convergence, prior routes, and complete QC/cause records. Optional array/network stacks remain diagnostic and do not replace per-detector products.

5 Conformance evaluation sequence

E1. Bind immutable observation, parent, source, nominal-beam, detector, and contract identities; resolve the raw $\Delta f/f$ convention.

E2. Admit the standardized detector map only after causal validity, support, WCS/metric, response, and covariance checks.

E3. Resolve requested/effective/realized source, background, prior, fit, iteration, calibration, sensitivity, and QC policy without hidden defaults.

E4. Locate candidates with compatible soft prior plus blind fallback; compare admitted candidates under one measurement objective.

E5. Fit Equation 9 jointly in the physical coordinate plane; resolve model Jacobian, covariance, identifiability, residual, adequacy, and convergence.

E6. Derive raw/horizon geometry using the declared centroid sign/frame transformation and parent-sample contribution support, then derive conditional FWHM/area and compatible broadening while preserving origin/pivot limitations.

E7. Evaluate and accept `flxscale` under BEAM's source-atmosphere authority; separately classify and summarize off-source scans and publish strictly positive `sens` only when its open policy inputs are resolved.

E8. Populate independent quantity states for every attempted detector, publish the mandatory APT and dense companions atomically, and retain all unavailable causes and downstream-use limitations.

6 Required evidence classes

Evidence layer	What it may establish
Algebraic	identities, units, invariants, coordinate/tensor transformations, atmosphere ownership/composition, and state-machine logic
Implementation conformance	that a candidate implementation realizes every applicable requirement and prediction with no hidden substitution

Representation/response fidelity	that WCS, map response, support, source/nominal beam, atmosphere, covariance, and scan classification represent the intended observation
Observational performance	bias, scatter, uncertainty calibration, wing detectability, repeatability, and transfer over relevant observations
Science impact	separate accuracy needs and qualification for health, amplitude, calibration, sensitivity, Gaussian/empirical kernels, and complete-PSF claims
Production readiness	owner-approved numerical policies, operations, monitoring, failure handling, and evidence over the declared domain

The required future program includes PSF recovery across nominal, broadened, rotated, asymmetric, halo, multi-scale, cropped, masked, low-S/N, correlated- noise, background, and map-response cases; hidden-response/map-radius studies; kernel-impact propagation; coordinate/WCS/Jacobian and derivative checks; broadening and circular-limit tests; common-origin and conventional-pivot tests; per-detector angle and same-APT pointing-transfer tests. Authorship claims none of these results.

7 Open scientific-owner decision groups

These stable group IDs are the document-facing index into the nine atomic open questions in the external scientific-owner decision ledger.

Group ID	Subject	Atomic ledger questions
`SCI-BEAM-OD-001`	Sensitivity estimator and support	`SCI-BEAM-ODQ-101--103`
`SCI-BEAM-OD-002`	Adequacy, wings, and kernels	`SCI-BEAM-ODQ-104--106`, `109`
`SCI-BEAM-OD-003`	Physical pivot registration	`SCI-BEAM-ODQ-107--108`

8 Normative notation

Symbol	Meaning
o, a, d, p, s, t	Observation, array, stable detector occurrence, Beammap pixel, scan, and time/sample coordinate. In-memory indices are zero-based; persisted conventions remain explicitly declared.
$x_d(t) = \Delta f_d(t)/f_d$	Raw, uncalibrated fractional KID resonance-frequency shift for detector d. "Raw" names the physical observable, not necessarily untouched acquisition bytes.
$\mathbf{x}(t) = \{x_d(t)\}_d$; `xs`	Detector collection of raw $\Delta f/f$ signals. Neither symbol denotes a calibrated mJy beam^{-1} signal.
$x_{d,p}^{\mathrm{BM}}$	Standardized per-detector Beammap datum fitted by v0.1, in $\Delta f/f$, with declared conditioning, support, mapmaking, WCS, and response lineage.
(r_p, c_p); $\boldsymbol{\vartheta}_p = (\vartheta_{x,p}, \vartheta_{y,p})^{\mathsf{T}}$	Map pixel coordinate and its metrically orthonormal angular tangent-plane coordinate in arcseconds.
f_{WCS}; $\mathbf{J}_{\mathrm{sky}}$	Full pixel-to-tangent-plane transformation and local coordinate Jacobian $\partial\boldsymbol{\vartheta}/\partial(r, c)$, including scale, signs, handedness, rotation, shear, and azimuth metric.
$\mathcal{I}_d$	Admitted per-detector map support after upstream validity, finite-data, WCS, and effective fit-support rules.
$\mathbf{C}_d^{\mathrm{BM}}$	Declared covariance of admitted Beammap pixels, or a typed approximation $\widetilde{\mathbf{C}}_d^{\mathrm{BM}}$.
$S(\mathbf{u}; \boldsymbol{\phi})$	Declared non-negative calibrator surface-brightness shape with unit integral, finite first moment, zero-first-moment reference origin, nuisance parameters $\boldsymbol{\phi}$, and uncertainty.
$\boldsymbol{\Sigma}_{\mathrm{eff},d}$	Symmetric positive-definite tensor of the observation-local effective PSF core measured in the standardized detector Beammap, in arcsec^2.
$G(\mathbf{u}; \boldsymbol{\Sigma})$	Peak-normalized elliptical Gaussian core associated with tensor $\boldsymbol{\Sigma}$.
$H = S * G$; $T = H/H(\mathbf{0})$	Source-convolved profile and template normalized to unity at the declared source-reference origin. That origin need not be the profile maximum.
A_d^{obs}	Fitted observed-plane source excess at the translated reference origin, in $\Delta f/f$.
$\boldsymbol{\mu}_{d,\mathrm{fit}}$; $\mathbf{S}_{\mu\to x,o}$	Fitted source-reference centroid and the declared sign/frame transformation from fitted source centroid to raw relative detector coordinate for observation o.
$(x_t, y_t)_{d,\mathrm{raw}}$; $(x_t, y_t)_d$	Raw observation-local relative detector coordinate and its horizon-derotated relative focal-plane coordinate, in arcseconds.
$w_{d,p,t}^{\mathrm{BM}}$; $\chi_{d,\mathrm{eff}}$	Parent-sample contribution weight to detector-map pixel p and the effective field-derotation angle derived from those parent weights after restriction through the realized fit support.
$\boldsymbol{\Sigma}_{\mathrm{nom},a}$; $\boldsymbol{\Sigma}_{\mathrm{broad},d}$	Fixed versioned nominal-beam tensor and compatible difference $\boldsymbol{\Sigma}_{\mathrm{eff},d} - \boldsymbol{\Sigma}_{\mathrm{nom},a}$.
$\sigma_{\mathrm{maj}}, \sigma_{\mathrm{min}}, \theta$	Effective-core major/minor Gaussian standard deviations and orientation, with ordered positive axes, period π, and unavailable orientation at circularity.
$\mathbf{q}_d$; $\mathbf{J}_d^{\mathrm{fit}}$; $\mathbf{V}_{q,d}$	Complete realized map-model free-parameter vector, local fit Jacobian, and joint map-fit parameter covariance.
$\boldsymbol{\eta}_d$; $\mathbf{J}_d^{\mathrm{drv}}$; $\mathbf{V}_{\mathrm{drv},d}$	External calibration/sensitivity nuisance vector, separate derived-product Jacobian, and propagated covariance retaining material dependence between fit and derived stages.
$\boldsymbol{\beta}_d$; $r_{d,p}$	Declared constant or planar background coefficients and admitted residual $x_{d,p}^{\mathrm{BM}} - m_{d,p}$.
$F_{s,a}^{\mathrm{TOA,nom}}$	Required calibrator source flux at top of atmosphere, in mJy beam$_{\mathrm{nom}}^{-1}$ for the fixed nominal beam, at the same source-reference origin as A_d^{obs} and under the declared epoch/passband/spectral/finite-source convention.
$C_{s,a}^{\mathrm{eff}}$; $T_{s,a}^{\mathrm{eff}}$	Effective source-observation atmospheric correction and transmission, with $C^{\mathrm{eff}} = 1/T^{\mathrm{eff}}$, matched to the amplitude support and weighting.
$A_d^{\mathrm{TOA-eq}}$; flxscale$_d$	Top-of-atmosphere-equivalent fitted amplitude and BEAM-published conversion factor $F_{s,a}^{\mathrm{TOA,nom}}/A_d^{\mathrm{TOA-eq}}$ in mJy beam$_{\mathrm{nom}}^{-1}/(\Delta f/f)$.
$\mathcal{S}_{d,\mathrm{off}}$; $n_{d,s}$; sens$_d$	Detector-specific admitted off-source scans, non-negative declared time-normalized scan noise statistic, and finite strictly positive BEAM-published NEFD-like sensitivity in mJy beam$_{\mathrm{nom}}^{-1}\sqrt{\mathrm{s}}$ when available.
$\mathcal{A}_d^{(k)}, c_d^{(k)}, \mathcal{V}^{(k)}$	Parameter-availability set, selected locator/measurement-candidate identity, and valid-detector set at iteration k.

The words *shall* and *shall not* are normative. *May* marks an allowed option whose realized state must be recorded. "Unavailable" is a typed scientific state with one or more causes, never an undocumented numeric sentinel.

9 Definitions and scientific state model

SCI-BEAM v0.1 operation. SCI-BEAM consumes standardized per-detector Beammaps in raw $\Delta f/f$, fits an observation-local effective PSF core and source amplitude, derives raw and horizon-derotated relative detector coordinates, derives top-of-atmosphere nominal-beam `flxscale` and NEFD-like `sens`, and publishes the mandatory Beammap APT with its empirical maps, diagnostics, covariance, state, and lineage companions.

Raw detector signal and standardized Beammap. $x_d(t) = \Delta f_d(t)/f_d$ is the uncalibrated physical detector observable; `xs` denotes its detector collection. A standardized detector Beammap is a map-domain representation of that same observable with declared reference frequency, sign, zero/baseline treatment, conditioning, support, mapmaking, pixelization, WCS, filtering, response, and covariance state. A map already in mJy beam^{-1} is not a v0.1 calibration-estimation input.

Primary estimand. The fitted shape is the observation-local effective PSF core measured in the standardized per-detector Beammap. The elliptical Gaussian is a compact summary of the measurable high-S/N core after all response embodied in the parent map. It is not automatically the intrinsic detector-plus-telescope optical beam and does not establish the complete PSF outside the finite, noise-limited field.

Empirical response and completeness. The empirical Beammap is the finite-field measurement. The fit, residual map, support, background, map extent, covariance state, and response lineage are required companions. Gaussian-core validity, measured wing constraint, and complete-PSF interpretation are separate states. A non-detection of broad response is not evidence of absence; diffuse power may be below noise, outside the field, or background-degenerate.

Finite-source and amplitude convention. The source shape has unit integral and a declared zero-first-moment reference origin. The effective Gaussian is convolved with that shape and the resulting template is normalized to unity at the same origin. A_d^{obs} is the fitted observed-plane excess at that origin in $\Delta f/f$; it is not necessarily a profile maximum. $F_{s,a}^{\mathrm{TOA,nom}}$ is the required top-of-atmosphere calibrator source flux in mJy beam$_{\mathrm{nom}}^{-1}$ for the fixed nominal beam under the identical reference-origin convention. That declared value enters `flxscale`.

Effective PSF tensor. The scientific shape is one symmetric positive-definite two-dimensional tensor. Major/minor/orientation, independent tensor elements, Cholesky factors, or an equivalent positive-definite fit parameterization may realize it, but all three independent degrees of freedom are jointly fitted. Axis-aligned-only fits, separate one-dimensional fits, and post-hoc rotation estimates are nonconformant. At circularity the invariant tensor remains available and orientation is unavailable.

Widths and Gaussian area. Major/minor FWHM are compact representations of the effective Gaussian core. $2\pi\sigma_{\mathrm{maj}}\sigma_{\mathrm{min}}$ is available only as the solid angle of that Gaussian core under an explicit Gaussian-response interpretation. It is not the complete beam solid angle when material unmeasured response may exist.

Coordinate plane and Jacobian. The Euclidean ellipse is evaluated only in a metrically orthonormal angular tangent plane. The full WCS transform, or a validated local linearization with quantified error, establishes angular scale, signs, handedness, rotation, shear/cross terms, correct azimuth metric, and material spatial variation. Raw azimuth/elevation offsets are insufficient unless this transformation establishes their orthonormality.

Raw and horizon coordinates. $(x_t, y_t)_{\mathrm{raw}}$ is the relative detector coordinate obtained from the fitted source centroid by the explicit declared sign/frame transformation $\mathbf{S}_{\mu\to x,o}$. (x_t, y_t) is the coordinate after derotation into the declared horizon-fixed focal-plane frame. The effective derotation angle is derived for each detector from the parent-sample time/elevation support and contribution weights that produced the fitted map pixels, propagated through the realized fit support; the PSF tensor transforms consistently.

Origin gauge and rotation pivot. Beammap APT coordinates are defined only up to a common translation. The deterministic recentering rule is recorded but has no physical boresight meaning. The physical field-rotation pivot is separately classified as established, estimated, conventional, or unavailable. A conventional pivot is not telescope truth. BEAM does not publish ensemble translation as pointing error, boresight offset, absolute array position, or pointing-model correction.

Same-APT pointing transfer. Bracketing pointing observations and the associated science observation use the same immutable APT artifact and AST rotation convention. A different artifact is invalid unless a separately authorized coordinate transform proves equivalence. Physical focal-plane-to-boresight registration is a future calibration product.

Broadening tensor. For compatible Gaussian and response conventions, $\mathbf{\Sigma}_{\text{broad}} = \mathbf{\Sigma}_{\text{eff}} - \mathbf{\Sigma}_{\text{nom}}$ is a telescope-health diagnostic. A literal Gaussian-smoothing interpretation requires positive-semidefinite support within uncertainty. An indefinite result is preserved as model-inadequacy or incompatibility evidence and is not forced into a kernel.

Fit and derived-product covariance. The local map-fit Jacobian and covariance cover the fitted and admitted map-model parameters. Separate derived-product Jacobians add the calibrator source flux, nominal-beam convention, source atmosphere, source-model systematics, scan-noise statistic, and exposure normalization for `flxscale` and `sens`, retaining material dependence between stages.

Top-of-atmosphere `flxscale`. BEAM owns source-observation atmosphere treatment, candidate evaluation, scientific acceptance, and publication. The factor converts raw $\Delta f/f$ to top-of-atmosphere mJy per fixed nominal beam. Source atmosphere belongs to its lineage; target- observation atmosphere is a later SCI-CAL operation. TolProj may associate or transform the published source APT under its own approved contract, and SCI-CAL consumes the selected observation-specific factor.

Sensitivity. `sens` is a per-detector, top-of-atmosphere, nominal-beam, NEFD-like point-source sensitivity. It is the product of the magnitude of `flxscale` and a declared non-negative, robust, time-normalized off-source scan noise statistic on the matching source-atmosphere convention. It is finite and strictly positive whenever available; the sign of `flxscale` remains separately preserved. It may support rough relative weights and health diagnosis; it is not calibration, total uncertainty, exact inverse variance, empirical significance, or a complete noise model.

Deprecated `responsivity`. `responsivity` is not a BEAM estimand, calibration factor, weight, validity field, transfer parameter, or active scientific authority. It is deprecated compatibility metadata only; its absence does not make the canonical APT incomplete and no value is fabricated.

Independent quantity state. Every attempted detector remains in the BEAM result. Empirical map, amplitude, centroid, raw coordinate, horizon coordinate, tensor, orientation, broadening, model adequacy, wing constraint, complete-PSF interpretation, `flxscale`, `sens`, health use, and kernel-use qualification each have independent availability, validity, uncertainty, and cause state. One failure does not erase unrelated valid quantities.

Canonical Beammap APT. The APT is a mandatory BEAM scientific output. Per-detector content resolves identity, raw/horizon coordinates and covariance, derotation, effective tensor/FWHM/orientation, amplitude, `flxscale`, `sens`, and independent states. APT-level lineage resolves observation, calibrator, nominal beam, atmosphere, raw-signal convention, WCS/Jacobian, pivot/origin, response, models, estimators, versions, and parents. Immutable links resolve dense empirical maps, supports, residuals, Jacobians, joint covariance, diagnostics, broadening, wing constraints, off-source scan records, convergence/prior records, and cause/QC records.

Authority layers. Algebraic correctness, implementation conformance, map/response representation fidelity, observational performance, science-impact qualification, and production readiness are separate evidence layers. No earlier layer implies a later one.

Compatibility. Compatibility requires explicit agreement of scientific contract, observation/source/detector identities, raw-signal definition, reference plane, units, nominal beam, finite-source convention, WCS/Jacobian/frame, origin/pivot/rotation convention, map response, support, covariance, atmosphere, validity, and intended use. Equal payloads, filenames, row positions, or local labels are insufficient.

10 Normative scientific model

10.1 Raw signal, coordinate metric, and map-domain fit

For detector d, the raw uncalibrated observable and detector collection are

$$x_d(t) \equiv \frac{\Delta f_d(t)}{f_d}, \qquad \mathbf{x}(t) = \{x_d(t)\}_d. \tag{1}$$

The standardized detector Beammap $x^{\mathrm{BM}}_{d,p}$ remains in $\Delta f/f$. Its physical fit coordinate is

$$\boldsymbol{\vartheta}_p = f_{\mathrm{WCS}}(r_p, c_p), \qquad \mathbf{J}_{\mathrm{sky},p} = \left.\frac{\partial(\vartheta_x, \vartheta_y)}{\partial(r, c)}\right|_{(r_p, c_p)}. \tag{2}$$

The full transformation shall be evaluated per pixel unless a declared local linearization has bounded error over the complete fit region.

Let

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \tag{3}$$

$$\boldsymbol{\Sigma}_{\mathrm{eff}} = \mathbf{R}(\theta)\,\mathrm{diag}(\sigma^2_{\mathrm{maj}}, \sigma^2_{\mathrm{min}})\,\mathbf{R}(\theta)^{\mathsf{T}}, \tag{4}$$

$$G(\mathbf{u}; \boldsymbol{\Sigma}_{\mathrm{eff}}) = \exp\left(-\tfrac{1}{2}\mathbf{u}^{\mathsf{T}}\boldsymbol{\Sigma}_{\mathrm{eff}}^{-1}\mathbf{u}\right), \tag{5}$$

$$H(\mathbf{u}; \boldsymbol{\Sigma}_{\mathrm{eff}}, \boldsymbol{\phi}) = \int S(\mathbf{t}; \boldsymbol{\phi}) G(\mathbf{u} - \mathbf{t}; \boldsymbol{\Sigma}_{\mathrm{eff}})\, d^2\mathbf{t}, \tag{6}$$

$$T(\mathbf{u}; \boldsymbol{\Sigma}_{\mathrm{eff}}, \boldsymbol{\phi}) = \frac{H(\mathbf{u}; \boldsymbol{\Sigma}_{\mathrm{eff}}, \boldsymbol{\phi})}{H(\mathbf{0}; \boldsymbol{\Sigma}_{\mathrm{eff}}, \boldsymbol{\phi})}. \tag{7}$$

The admitted source model shall satisfy

$$S \geq 0, \qquad \int S(\mathbf{u}; \boldsymbol{\phi}) d^2\mathbf{u} = 1, \qquad \int \mathbf{u} S(\mathbf{u}; \boldsymbol{\phi}) d^2\mathbf{u} = \mathbf{0}, \qquad H(\mathbf{0}) > 0. \tag{8}$$

Thus $T(\mathbf{0}) = 1$ at the declared reference origin, which need not be the global maximum for an asymmetric source.

The canonical v0.1 map-domain model is

$$x^{\mathrm{BM}}_{d,p} = A^{\mathrm{obs}}_d T\left(\boldsymbol{\vartheta}_p - \boldsymbol{\mu}_{d,\mathrm{fit}}; \boldsymbol{\Sigma}_{\mathrm{eff},d}, \boldsymbol{\phi}\right) + \mathbf{b}_p^{\mathsf{T}}\boldsymbol{\beta}_d + r_{d,p}. \tag{9}$$

The effective tensor summarizes the core already embodied in the standardized map response. Equation 9 is not a deconvolution of mapmaking, pixelization, filtering, finite support, or any other parent response.

For admitted map covariance, the estimator minimizes

$$Q_d = (\mathbf{x}^{\mathrm{BM}}_d - \mathbf{m}_d)^{\mathsf{T}}(\mathbf{C}^{\mathrm{BM}}_d)^{-1}(\mathbf{x}^{\mathrm{BM}}_d - \mathbf{m}_d) + \log|\mathbf{C}^{\mathrm{BM}}_d|, \tag{10}$$

up to additive constants. A declared approximation $\widetilde{\mathbf{C}}^{\mathrm{BM}}_d$ may replace $\mathbf{C}^{\mathrm{BM}}_d$ only with conditional labels. Singular covariance requires a declared retained- subspace or regularization procedure; silent replacement is forbidden.

10.2 Model Jacobian, covariance, and derived shape

For every realized free parameter q_j, including amplitude, centroid, all three independent tensor degrees of freedom, background, and admitted source nuisance parameters, the fit resolves

$$J^{\mathrm{fit}}_{ij} = \frac{\partial m_i}{\partial q_j}. \tag{11}$$

Analytic, automatic, or verified numerical differentiation may realize the Jacobian. A local likelihood covariance may be reported as

$$\mathbf{V}_{q,d} = \left[\tfrac{1}{2}\nabla\nabla^{\mathsf{T}} Q_d(\widehat{\mathbf{q}}_d)\right]^{-1} \tag{12}$$

only for an identifiable interior solution under the declared likelihood and covariance model. The map-fit covariance retains material width–orientation, centroid–shape, amplitude–width, background–centroid, source–beam, and coordinate–shape terms among the fitted and admitted map-model parameters.

For calibration and sensitivity, let $\boldsymbol{\eta}_d$ contain the external calibrator source flux, nominal-beam convention, source atmosphere, source-model systematic terms, scan-noise statistic, exposure normalization, and other admitted derived-stage inputs. A separate derived-product Jacobian $\mathbf{J}^{\mathrm{drv}}_{(q,\eta)}$ propagates the joint fit/input covariance:

$$\mathbf{V}_{\mathrm{drv},d} = \mathbf{J}^{\mathrm{drv}}_{(q,\eta)} \mathbf{V}_{(q,\eta)} \left(\mathbf{J}^{\mathrm{drv}}_{(q,\eta)}\right)^{\mathsf{T}}. \tag{13}$$

Material dependence between fit and derived stages is retained. Omitted material terms are named and prohibit a “total” label. At circularity shape covariance is expressed in an invariant tensor basis and angle uncertainty is unavailable.

The compact Gaussian-core quantities are

$$\mathrm{FWHM}_{\mathrm{maj,min}} = 2\sqrt{2\ln 2}\,\sigma_{\mathrm{maj,min}}, \qquad \Omega_G = 2\pi\sigma_{\mathrm{maj}}\sigma_{\mathrm{min}}. \tag{14}$$

Ω_G is available only as the fitted Gaussian-core solid angle under a declared interpretation; it is not automatically the complete beam solid angle.

For compatible nominal/effective Gaussian and response conventions,

$$\boldsymbol{\Sigma}_{\mathrm{broad},d} = \boldsymbol{\Sigma}_{\mathrm{eff},d} - \boldsymbol{\Sigma}_{\mathrm{nom},a(d)}. \tag{15}$$

A literal Gaussian-broadening interpretation requires the difference to be positive semidefinite within propagated uncertainty. Otherwise the tensor is unavailable as a smoothing kernel and the indefinite result remains model-inadequacy or convention-incompatibility evidence.

10.3 Relative coordinates, derotation, and pivot status

Let $\mathbf{r}_{od}$ denote the fitted relative coordinate in the observation-local plane and $\mathbf{u}_d$ the horizon-fixed relative coordinate. The exact fitted-centroid relation is

$$\mathbf{r}_{od} = \mathbf{S}_{\mu\to x,o}\boldsymbol{\mu}_{d,\mathrm{fit}}, \tag{16}$$

where the declared transformation supplies the source-centroid to detector-coordinate sign and frame convention. It shall not be inferred from implementation behavior. The physical rotation relation is affine,

$$\mathbf{r}_{od} = \mathbf{R}(\chi_{od})\mathbf{u}_d + \mathbf{t}_o(\chi_{od}), \tag{17}$$

where the common translation contains the arbitrary coordinate-origin gauge and any unresolved pivot/pointing term. The v0.1 derotated product uses the declared conventional transform

$$\mathbf{u}_d = \mathbf{R}(-\chi_{d,\mathrm{eff}})\left(\mathbf{r}_{od} - \mathbf{o}_{\mathrm{conv}}\right), \qquad \boldsymbol{\Sigma}_{d,\mathrm{hor}} = \mathbf{R}(-\chi_{d,\mathrm{eff}})\boldsymbol{\Sigma}_{d,\mathrm{raw}}\mathbf{R}(-\chi_{d,\mathrm{eff}})^{\mathsf{T}}, \tag{18}$$

with deterministic common recentering. $\mathbf{o}_{\mathrm{conv}}$ is not a physical boresight unless separately established. $\chi_{d,\mathrm{eff}}$ is derived from the parent-sample time/elevation support and contribution weights $w^{\mathrm{BM}}_{d,p,t}$ that formed pixels $p \in \mathcal{I}_d$, propagated through the realized pixel-domain fit support. Pixel fit weights alone do not resolve parent timing. A single observation-average angle is allowed only with negligible validated error.

10.4 Top-of-atmosphere calibration and sensitivity

The source model supplies $F^{\mathrm{TOA,nom}}_{s,a}$ in $\mathrm{mJy\,beam}^{-1}_{\mathrm{nom}}$ for the fixed nominal beam at the same reference origin as A^{obs}_d. With source-observation correction $C^{\mathrm{eff}}_{s,a} = 1/T^{\mathrm{eff}}_{s,a}$,

$$A^{\mathrm{TOA-eq}}_d = C^{\mathrm{eff}}_{s,a} A^{\mathrm{obs}}_d, \tag{19}$$

$$\mathrm{flxscale}_d = \frac{F^{\mathrm{TOA,nom}}_{s,a}}{A^{\mathrm{TOA-eq}}_d} = \frac{T^{\mathrm{eff}}_{s,a} F^{\mathrm{TOA,nom}}_{s,a}}{A^{\mathrm{obs}}_d}. \tag{20}$$

The factor has role and unit $[\mathrm{mJy\,beam}^{-1}_{\mathrm{nom}}/(\Delta f/f)]_{\mathrm{TOA}}$. Its uncertainty is obtained from the separate derived-product Jacobian in Equation 13, using the joint covariance of the map-fit amplitude and the nominal source flux, atmosphere, and other material derived-stage inputs. Source atmosphere appears here; target atmosphere remains outside SCI-BEAM.

For detector-specific off-source scan set $\mathcal{S}_{d,\mathrm{off}}$,

$$
\begin{aligned}
n_{d,s}^{\mathrm{obs}} &= \mathcal{N}[x_{d,s}(t)], && (21)\\
n_{d,s}^{\mathrm{TOA-eq}} &= C_{s,a,s}^{\mathrm{eff}}\, n_{d,s}^{\mathrm{obs}}, && (22)\\
n_{d}^{\mathrm{off}} &= \mathrm{median}_{s\in\mathcal{S}_{d,\mathrm{off}}}\, n_{d,s}^{\mathrm{TOA-eq}}, && (23)\\
\mathrm{sens}_d &= |\mathrm{flxscale}_d|\, n_d^{\mathrm{off}}. && (24)
\end{aligned}
$$

$\mathcal{N}$ includes a declared time/exposure normalization so n_d^{off} has $\Delta f/f\sqrt{\mathrm{s}}$ and `sens` has $\mathrm{mJy\,beam_{nom}^{-1}}\sqrt{\mathrm{s}}$. An available `sens` is finite and strictly positive. The physical sign of `flxscale` is retained in the calibration record and only its magnitude enters Equation 24. The exact source exclusion, scan admission, baseline, frequency/bandwidth, robust statistic, correlated- noise, minimum-count, exposure, partial-scan, and uncertainty rules remain owner-controlled. The retained scan scatter is a required stability or uncertainty diagnostic. Approximate downstream weights may satisfy $\omega_d \propto \mathrm{sens}_d^{-2}$ only under a separately governed weighting policy.

10.5 Observation-local convergence

An observation-local iteration may alternate locator and measurement phases. The effective policy declares per-parameter metrics d_j and scales s_j, objective scale s_Q, tolerances, support and valid-detector stability rules, a consecutive-transition count, and $N_{\max}$. Orientation uses

$$d_\pi(a,b) = \min_{n\in\mathbb{Z}} |a - b + n\pi|.$$

With parameter-availability set $\mathcal{A}_d^{(k)}$, selected candidate $c_d^{(k)}$, support $\mathcal{I}_d^{(k)}$, and valid-detector set $\mathcal{V}^{(k)}$, convergence requires

$$
\begin{aligned}
\max_{j\in\mathcal{A}_d^{(k)}\cap\mathcal{A}_d^{(k-1)}} \frac{d_j(q_j^{(k)}, q_j^{(k-1)})}{s_j} &\le \epsilon_j, \qquad \mathcal{A}_d^{(k)} = \mathcal{A}_d^{(k-1)},\\
\frac{|Q^{(k)} - Q^{(k-1)}|}{s_Q} &\le \epsilon_Q, \qquad c_d^{(k)} = c_d^{(k-1)},\\
\mathcal{I}_d^{(k)} = \mathcal{I}_d^{(k-1)}, &\quad \mathcal{V}^{(k)} = \mathcal{V}^{(k-1)}.
\end{aligned} \tag{25}
$$

for the declared number of consecutive transitions. Availability changes fail stability; a parameter unavailable at both transitions contributes no numeric metric. Optimizer completion alone is never scientific convergence, and v0.1 defines no universal numeric defaults.

11 Assumptions, dependencies, and unavailable claims

A1. The raw detector observable is $x_d = \Delta f_d / f_d$. Its exact reference frequency, sign, zero/baseline treatment, conditioning, support, and valid loading/tune/signal domain are declared inputs rather than inferred by BEAM.

A2. The primary v0.1 fit input is a standardized detector-resolved Beammap in $\Delta f/f$. A calibrated map is not the calibration-estimation input, and a timestream fit is not the v0.1 production estimator.

A3. Parent mapmaking, pixelization, projection, gridding, filtering, finite boundary, WCS, response, validity, and covariance provenance are supplied and preserved. The fitted tensor is conditioned on that map response.

A4. The coordinate transformation establishes a metrically orthonormal angular plane and its uncertainty. A local Jacobian is used only with validated approximation error over the fit region.

A5. The immutable calibrator identity and source model are supplied for the declared epoch, passband, spectral, finite-source, and reference-origin convention, with uncertainty or explicit unavailable state.

A6. $F^{\mathrm{TOA,nom}}_{s,a}$ is supplied as the top-of-atmosphere calibrator source flux in $\mathrm{mJy\,beam}^{-1}_{\mathrm{nom}}$ for the fixed versioned nominal beam at the fitted reference origin.

A7. The effective Gaussian family is only a compact core model. A good fit does not prove the complete PSF, absence of wings, intrinsic optical response, or adequacy for a downstream kernel.

A8. The admitted support and constant/planar background jointly identify the reported source, centroid, three tensor degrees of freedom, amplitude, source nuisance, and background parameters. Unidentified quantities are unavailable.

A9. The covariance or declared approximation describes the retained map support sufficiently for the statistical claim. A diagonal approximation does not authorize calibrated uncertainty under correlated residuals.

A10. The full local map-model Jacobian and fit covariance preserve material cross terms among fitted and admitted map-model parameters. Separate derived-product Jacobians propagate external calibration and sensitivity inputs while retaining material dependence between stages. Missing nuisance covariance is unavailable, not independent or zero.

A11. The nominal and effective tensors share compatible frame, Gaussian, source, and response conventions before broadening is formed. Positive- semidefinite support is required for a literal Gaussian smoothing meaning.

A12. The centroid-to-detector sign/frame transformation, horizon derotation law, parent-sample support and contribution weights, effective per-detector angle, fit support, conventional pivot, origin gauge, and uncertainty are supplied. Pixel-domain fit weights alone are not parent timing evidence. Neither the conventional origin nor ensemble translation is physical boresight truth.

A13. Bracketing pointing observations and their associated science observation use the same immutable APT artifact and AST rotation convention unless a separately authorized transformation proves equivalence.

A14. BEAM owns source-observation atmosphere used in `flxscale`; the operator and support match the fitted amplitude. Target-observation atmosphere is excluded and later applied by SCI-CAL.

A15. The scan-domain `sens` structure is fixed and uses |`flxscale`| so an available sensitivity is finite and strictly positive. Its numerical noise, source-exclusion, admission, exposure, bandwidth, and uncertainty policies remain open owner decisions. No default is inferred.

A16. A compatible TolAPT prior is soft locator initialization and bounded gating only. It contributes no measurement-objective penalty, is not exact detector or position truth, cannot veto the blind route, and records influence.

A17. Every attempted detector and per-quantity state is published atomically with the mandatory APT and resolvable dense companions. One unavailable field does not erase independently valid fields.

A18. Numerical QC and production thresholds are versioned evidence-based policy, not physical constants derived from this model.

Dependencies on calibrator/source modeling, nominal-beam authority, source atmosphere, ALIGN/AST, mapmaking/response, TolAPT priors, TolProj child-APT transformation, SCI-CAL target calibration, weighting, and downstream kernel science remain explicit. This contract defines their interfaces but does not certify their inputs, implementations, or validation evidence.

12 Normative requirements

ID	Requirement
SCI-BEAM-REQ-001	The result shall bind immutable observation, source, array, detector occurrence, parent map, contract/document, and requested/effective/realized policy identities.
SCI-BEAM-REQ-002	The raw observable shall be $x_d = \Delta f_d / f_d$ and `xs` shall mean its detector collection; reference frequency, sign, zero/baseline, conditioning, support, and valid operating domain shall be explicit.
SCI-BEAM-REQ-003	The primary v0.1 fit input shall be a standardized detector-resolved Beammap in $\Delta f / f$ with stable detector/observation binding; calibrated maps and timestream-domain fits shall not substitute for it.
SCI-BEAM-REQ-004	The parent Beammap shall carry complete mapmaking, pixelization, projection, gridding, filtering, response, support, validity, covariance, and provenance state.
SCI-BEAM-REQ-005	Equation 2 shall establish a metrically orthonormal angular tangent plane, including scale, signs, handedness, rotation, shear/cross terms, azimuth metric, uncertainty, and material spatial variation.
SCI-BEAM-REQ-006	The source input shall declare immutable identity, epoch, passband/spectral convention, finite extent, zero-first-moment reference origin, normalization, uncertainty, and upstream authority.
SCI-BEAM-REQ-007	The fixed nominal-beam source prediction shall supply $F_{s,a}^{\mathrm{TOA,nom}}$ in top-of-atmosphere mJy $\mathrm{beam}_{\mathrm{nom}}^{-1}$ at the fitted reference origin, plus nominal-beam identity/tensor and uncertainty.
SCI-BEAM-REQ-008	Upstream-invalid pixels or detectors shall be excluded before payload evaluation; missing, disabled, rejected, invalid, failed, and unavailable shall remain distinct causes.
SCI-BEAM-REQ-009	The v0.1 forward model shall be Equation 9: one finite-source-convolved, full 2-D effective Gaussian core plus a declared constant or planar background on admitted map support.
SCI-BEAM-REQ-010	The realized shape fit shall span all three independent symmetric positive-definite tensor degrees of freedom; axis-aligned-only, separate one-dimensional, or post-hoc orientation fits are nonconformant.
SCI-BEAM-REQ-011	The ellipse shall use ordered positive axes and period-π orientation in the declared frame; at circularity the tensor remains available and orientation is unavailable.
SCI-BEAM-REQ-012	The finite-source template shall satisfy Equations 7–8; $T(\mathbf{0}) = 1$ and the reference-origin convention/model identity shall accompany every amplitude.
SCI-BEAM-REQ-013	A_d^{obs} shall mean observed-plane $\Delta f / f$ excess at the translated source-reference origin; it shall not be called integrated flux or profile peak unless separately established.
SCI-BEAM-REQ-014	The background family, origin, coefficients, and uncertainty shall be recorded; unsupported adaptive or higher-order backgrounds shall not be introduced.
SCI-BEAM-REQ-015	Admitted fit support shall record pixel selection, exclusions/causes, connectedness, edge/crop state, masked regions, map extent, retained count, and the support used by every derived quantity.
SCI-BEAM-REQ-016	The objective shall be Equation 10 with declared covariance or typed approximation; silent weighting, retained-subspace choice, or regularization is forbidden.
SCI-BEAM-REQ-017	Residuals shall mean admitted map data minus the complete realized source-plus-background model and shall be published as a dense companion on their valid support.
SCI-BEAM-REQ-018	Local identifiability shall be assessed jointly for source, centroid, tensor, amplitude, background, and admitted nuisance parameters; non-identifiable quantities shall be unavailable with causes.
SCI-BEAM-REQ-019	The complete local model Jacobian of Equation 11 shall resolve every realized map-model free parameter, including coordinate/WCS nuisance parameters when jointly admitted, and its derivative method shall be validated against controlled perturbations.
SCI-BEAM-REQ-020	The fit shall produce or resolve a joint map-fit covariance satisfying Equation 12 where its assumptions hold and retaining material cross terms among fitted and admitted map-model parameters; omitted terms shall be named.
SCI-BEAM-REQ-021	Covariance from an approximation shall be labeled conditional and shall not be described as calibrated by algebra; at circularity shape covariance shall use an invariant tensor basis.
SCI-BEAM-REQ-022	The fitted tensor shall be labeled the observation-local effective Beammap PSF core; intrinsic optical beam, complete PSF, wing completeness, and map-response removal shall remain independently unavailable unless separately established.
SCI-BEAM-REQ-023	The empirical Beammap, valid-support map, extent, noise/covariance state, fitted model, residual map, background, adequacy diagnostics, and parent response provenance shall be required companions.

ID	Requirement
SCI-BEAM-REQ-024	A compatible TolAPT prior may initialize and bound locator gating but shall not enter the measurement objective as hidden truth, exact detector/position identity, or unconditional veto.
SCI-BEAM-REQ-025	Missing, incompatible, weak, or unsuccessful prior use shall invoke the declared blind locator route; admitted candidates shall be compared under one realized measurement objective.
SCI-BEAM-REQ-026	Prior producer/version, compatibility, route, displacement, gate influence, fallback, and selected-candidate origin shall be retained.
SCI-BEAM-REQ-027	Locator/measurement phases, parameter availability, objective, selected candidate, support, detector set, model identity, and their changes shall be one observation-local estimator trace.
SCI-BEAM-REQ-028	Convergence shall satisfy every separately declared component of Equation 25, including modulo-π angle distance; optimizer completion or maximum iteration alone shall not imply convergence.
SCI-BEAM-REQ-029	Every attempted detector shall remain represented and empirical map, amplitude, centroid, raw/horizon coordinates, tensor, angle, broadening, adequacy, wing/complete-PSF state, **`flxscale`**, **`sens`**, health use, and kernel use shall have independent availability/validity/cause state.
SCI-BEAM-REQ-030	Parameter validity shall require finite values where defined, positive-definite effective tensor, valid template normalization, declared-domain parameters, causal support, and finite strictly positive available **`sens`**; one invalid derived use shall not erase unrelated valid quantities.
SCI-BEAM-REQ-031	QC shall separate algebraic/model validity, empirical review cues, model inadequacy, complete-PSF/wing completeness, telescope-health use, calibration use, sensitivity use, and kernel-use qualification.
SCI-BEAM-REQ-032	Any numerical fit, support, adequacy, S/N, sensitivity, convergence, wing, or production threshold shall be a versioned evidence-based policy with owner and requested/effective/realized values; v0.1 defines no universal defaults.
SCI-BEAM-REQ-033	Quantitative adequacy diagnostics shall address structured residuals, support dependence, peak displacement, asymmetry, excess outside the core, multiple peaks/lobes, core/wing incompleteness, and possible amplitude bias; non-detection shall not prove absence of broad response.
SCI-BEAM-REQ-034	Major/minor FWHM and conditional Gaussian-core solid angle shall satisfy Equation 14; Gaussian-core area shall not be relabeled complete beam solid angle.
SCI-BEAM-REQ-035	For compatible conventions, broadening shall use Equation 15 with propagated covariance; literal Gaussian smoothing requires positive-semidefinite support, while an indefinite difference remains a typed inadequacy/incompatibility result.
SCI-BEAM-REQ-036	Raw detector coordinates shall be related to fitted source centroids by the explicit sign/frame transformation in Equation 16; raw and horizon coordinates, parent-sample support/contribution weights propagated through fit support, effective per-detector angle/range, rotation convention, transformation uncertainty, and consistently rotated PSF tensor shall be retained.
SCI-BEAM-REQ-037	The APT common-origin/recentering rule and pivot classification shall be explicit; conventional origin or ensemble translation shall not claim boresight, absolute array position, pointing error, or pointing-model correction.
SCI-BEAM-REQ-038	Bracketing pointing observations and the associated science observation shall use the same immutable APT artifact and AST rotation convention unless a separately authorized transform proves equivalence.
SCI-BEAM-REQ-039	BEAM shall derive, scientifically accept, and publish `flxscale` using Equation 20 from the required TOA source flux in mJy $\mathrm{beam}_{\mathrm{nom}}^{-1}$; source atmosphere belongs to BEAM and target atmosphere remains outside BEAM.
SCI-BEAM-REQ-040	`flxscale` shall carry its TOA nominal-beam unit/role, calibrator/epoch, source model, nominal beam, passband/spectrum, source-atmosphere operator/support, amplitude/support, detector/observation, acceptance state, and separate derived-product Jacobian/covariance retaining material fit-to-calibration dependence, uncertainty, and correlation scope.
SCI-BEAM-REQ-041	BEAM shall calculate and publish finite strictly positive NEFD-like `sens` with Equation 24 from |`flxscale`|, detector-specific trajectory-classified off-source scans, and the same TOA atmosphere/calibration convention; the signed `flxscale` shall remain preserved separately.
SCI-BEAM-REQ-042	The `sens` record shall identify scan noise/baseline/bandwidth/exposure/correlation/admission policy, source-exclusion rule, invalid/partial-scan handling, minimum support, retained scan values/scatter, limitations, and the separate derived-product Jacobian/covariance used for uncertainty; no exact numerical estimator is authorized until the owner resolves `SCI-BEAM-OD-001`.
SCI-BEAM-REQ-043	`responsivity` shall have no canonical BEAM role; its absence shall not invalidate the APT, legacy presence shall convey no authority, and new production shall not fabricate a value.
SCI-BEAM-REQ-044	The mandatory Beammap APT shall resolve the complete per-detector quantities and independent states, APT-level observation/calibration/geometry/response lineage, scientific/document versions, parent digests, and immutable links to all required dense companions, without prescribing current storage.
SCI-BEAM-REQ-045	Array/network/observation stacked PSFs may be optional diagnostics only with declared recentering, inclusion, weighting, normalization, coverage, detector variation, covariance, and parents; they shall not gate required per-detector results or become kernels without qualification.
SCI-BEAM-REQ-046	Outputs and validation plans shall separate algebraic correctness, implementation conformance, response representation fidelity, observational performance, science-impact/kernel qualification, and production readiness, and shall require the recovery, hidden-response, kernel-impact, geometry, rotation, same-APT, and perturbation studies named by this contract.

13 Falsifiable predictions and edge cases

These predictions test the contract, not current software or production readiness.

ID	Predicted invariant or state
SCI-BEAM-PRED-001	A product whose physical observable, reference frequency, sign, zero/baseline, conditioning, or unit differs from declared $\Delta f/f$ is scientifically incompatible even if its numeric array is identical.
SCI-BEAM-PRED-002	Replacing the standardized raw-signal detector Beammap with a calibrated map or a timestream fit changes the estimand and is rejected rather than producing a v0.1-equivalent result.
SCI-BEAM-PRED-003	Noiseless identifying data from an admitted circular effective core recover reference-origin A_d^{obs}, centroid, and common width with zero residual to numerical precision; tensor shape remains available and orientation is unavailable.
SCI-BEAM-PRED-004	Noiseless identifying data from a rotated non-circular core recover the complete tensor, ordered widths, and orientation modulo π; an axis-aligned-only realization leaves structured residual or biased shape.
SCI-BEAM-PRED-005	Axis-label exchange plus $\pi/2$ and orientation plus π leave the unconstrained ellipse invariant; canonicalization restores the same ordered tensor and angle in $[0, \pi)$.
SCI-BEAM-PRED-006	Full WCS transformation and an adequately validated local Jacobian agree within the declared approximation bound; omitting scale, azimuth metric, rotation, or shear biases centroid/tensor recovery in a controlled transformed injection.
SCI-BEAM-PRED-007	As admitted source extent tends to zero, the source-convolved normalized template tends continuously to the point-source effective Gaussian; for a finite asymmetric source the reference-origin amplitude remains defined without becoming a guaranteed profile maximum.
SCI-BEAM-PRED-008	A calibrator source flux not supplied in top-of-atmosphere $\mathrm{mJy\,beam}_{\mathrm{nom}}^{-1}$ for the declared fixed nominal beam and reference origin is incompatible with Equation 20, even when its numeric payload matches.
SCI-BEAM-PRED-009	Source-atmosphere correction represented by C^{eff} in the denominator or T^{eff} in the numerator gives the same Equation 20 result; source-atmosphere lineage belongs to BEAM and target atmosphere remains downstream.
SCI-BEAM-PRED-010	A valid effective-core amplitude/shape can coexist with unavailable `flxscale` when nominal-source or source-atmosphere authority is missing; the APT row and independent fit states remain present.
SCI-BEAM-PRED-011	For finite nonzero signed `flxscale` and positive admitted n_d^{off}, Equation 24 yields the same strictly positive `sens` for either `flxscale` sign; failed off-source support leaves `sens` unavailable without erasing calibration.
SCI-BEAM-PRED-012	Cropping, masked cores, finite radial extent, or low surface-brightness wings can preserve a precise Gaussian-core fit while wing/complete-PSF state remains unavailable; non-detection never becomes zero wing power.
SCI-BEAM-PRED-013	A planar background is recovered only on support identifying its gradients; broad response degenerate with the background produces support-dependent amplitude/shape or adequacy evidence rather than proof that broad response is absent.
SCI-BEAM-PRED-014	Analytic, automatic, and verified finite-difference model Jacobians agree within the declared perturbation tolerance on controlled cases; map-fit derivatives remain distinct from derived-product calibration/sensitivity Jacobians, and omission of a material derivative or cross-stage dependence is a conformance failure.
SCI-BEAM-PRED-015	Rotating coordinates transforms the tensor and joint covariance covariantly while leaving eigenvalues invariant; approaching circularity leaves invariant tensor covariance finite where identified and makes angle uncertainty unavailable.
SCI-BEAM-PRED-016	For injected compatible Gaussian broadening, Equation 15 recovers a positive-semidefinite tensor and its eigensystem; an indefinite difference is retained as inadequacy/incompatibility and is never clipped silently into a kernel.

`SCI-BEAM-PRED-017`	The declared centroid sign/frame transform plus parent-sample-contribution-weighted per-detector derotation recovers injected horizon-frame coordinates/tensors within declared uncertainty; reversing the centroid sign or substituting pixel fit weights or a material observation-average angle fails.
`SCI-BEAM-PRED-018`	Adding a common translation to every raw detector coordinate and applying the corresponding deterministic recentering leaves all relative geometry invariant; no absolute boresight claim changes because none is authorized.
`SCI-BEAM-PRED-019`	Conventional-pivot perturbations produce principally common elevation-dependent translation and, when detector effective angles differ, differential geometry; same-artifact/same-AST pointing transfer cancels the common gauge consistently, while mismatched immutable APT artifacts fail compatibility.
`SCI-BEAM-PRED-020`	A compatible correct soft prior may alter locator work but not the selected prior-free measurement optimum; a wrong prior loses to a stronger blind-route candidate under the common objective and records its influence.
`SCI-BEAM-PRED-021`	Optimizer completion with unstable parameter metrics, availability, objective, selected candidate, support, or detector set is non-converged; modulo-π angle equivalence and circular unavailability behave as Equation 25 declares.
`SCI-BEAM-PRED-022`	Removing deprecated `responsivity` from an otherwise complete canonical APT changes no BEAM estimand, validity, calibration, sensitivity, or required-product disposition.
`SCI-BEAM-PRED-023`	Independent state permits valid empirical map, tensor, amplitude, `flxscale`, or health diagnosis alongside other unavailable uses; every attempted detector remains in the mandatory APT and dense lineage.
`SCI-BEAM-PRED-024`	Synthetic or observational hidden wings, model-family mismatch, and downstream kernel perturbations can change amplitude, `flxscale`, `sens`, or science products despite finite core parameters; no kernel-use or production claim follows until the prescribed science-impact studies pass.

14 Minimum conformance report

A conformance report shall give, for every requirement and prediction, an evidence identifier, applicability and reason, realized scientific identity and policy, observed result, expected result, disposition, and unresolved dependency. It shall inventory required APT and companion products and shall not represent compilation, finite values, synthetic recovery, or a clean log as proof of a later evidence layer.